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Systematic Review

Systematic Review of Integrating Technology for Sustainable Agricultural Transitions: Ecuador, a Country with Agroecological Potential

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Abstract

Agroecology has traditionally been implemented using conventional methods. However, the integration of precision equipment, advanced methodologies, and digital technologies (DT) is now essential for transitioning to a more modern and efficient approach. While agroecological principles remain fundamental for planning and managing sustainable food systems by optimizing natural resources, technological tools can significantly support their implementation and adoption by farmers. This transition, however, must also consider socioeconomic factors and policy frameworks to ensure that technological advancements lead to meaningful improvements in farms and agroecosystems. Across both industrialized and emerging economies, various initiatives, such as precision agriculture, digital platforms, and e-commerce, are driving the digitalization of agroecology. These innovations offer clear benefits, including enhanced knowledge generation and direct improvements to the food supply chain; however, several barriers remain, including limited understanding of digital tools, high-energy demands, insufficient financial resources, economical constrains, weak policy support, lack of infrastructure, low digital learning by framers, etc. to facilitate the transition. This review looks for the understanding of how digitalization can align or conflict with local agroecological dynamics across distinct political frameworks and reality contexts because the information about DT adoption in agroecological practices is limited and it remains unclear if digital agriculture for scaling agroecology can considerably change power dynamics within the productive systems in regions of Europe and Latin America. In South America, among countries like Ecuador, with strong potential for agroecological development, where 60% of farms are less than 1 ha, and where farmers have expressed interest in agroecological practices, 80% have reported lacking sufficient information to make the transition to digitalization, making slow the adoption progress of these DT. While agroecology is gaining global recognition, its modernization through DT requires further research in technical, social, economic, cultural, and political dimensions to more guide the adoption of DT in agroecology with more certainty.

Keywords: agroecological principles; digitalization; Ecuador; precision agriculture; policy

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Academic Editor: Marko Vinceković

Received: 30 May 2025
Revised: 27 June 2025
Accepted: 30 June 2025
Published: 2 July 2025

Citation: Viera-Arroyo, W.; Binego, L.; Ryans, F.; López, D.; Moya, M.; Vera, L.; Caicedo, C. Systematic Review of Integrating Technology for Sustainable Agricultural Transitions: Ecuador, a Country with Agroecological Potential. *Sustainability* **2025**, *17*, 6053. https://doi.org/10.3390/su17136053

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1. Introduction

Agroecology is an interdisciplinary subject that implements ecological principles in the processes of agricultural systems [1]. The agroecological transition operates across multiple scales, ranging from individual farms to local communities and broader territories, and it is influenced by a complex interplay of social, economic, technological, cultural, political, and ecological factors [2]. This transition requires a deep understanding of agroecosystem functioning, including their structure and processes, the various modes of intervention to convert natural ecosystems into productive agroecosystems, and the external conditions that shape the feasibility of such transformations [2].

Agroecological principles are central to the design and management of sustainable food systems. They also contribute to building resilient agricultural systems that emulate natural ecological processes [3]. These principles aim to optimize nutrient cycling, organic matter turnover, soil biological activity, energy flows, water and soil conservation, and the balance between pests and their natural enemies [4]. By integrating ecological insights with socioeconomic considerations, agroecology offers a holistic framework for enhancing the resilience and productivity of food systems [1].

Despite its potential, the adoption of digital technologies (DT) in agriculture remains limited in many countries [5], and the interactions between digitalization and agroecological transitions are still not well understood [6]. The emerging concept of digital agroecology (DA) seeks to bridge this gap by combining agroecological principles with digital tools to support sustainable farming practices and food systems. This approach leverages data-driven technologies, such as Big Data, blockchain, IA, remote sensing, deep learning, robotics, autonomous systems, etc. (Figure 1), to facilitate agroecological transitions and address the challenges faced by farmers [7,8]. However, the integration of digital tools into agroecology remains a subject of debate. Advocates argue that digital farming can enhance environmental sustainability, while critics caution that it may conflict with the foundational principles of agroecology, particularly those emphasizing local knowledge, autonomy, and ecological harmony [9].

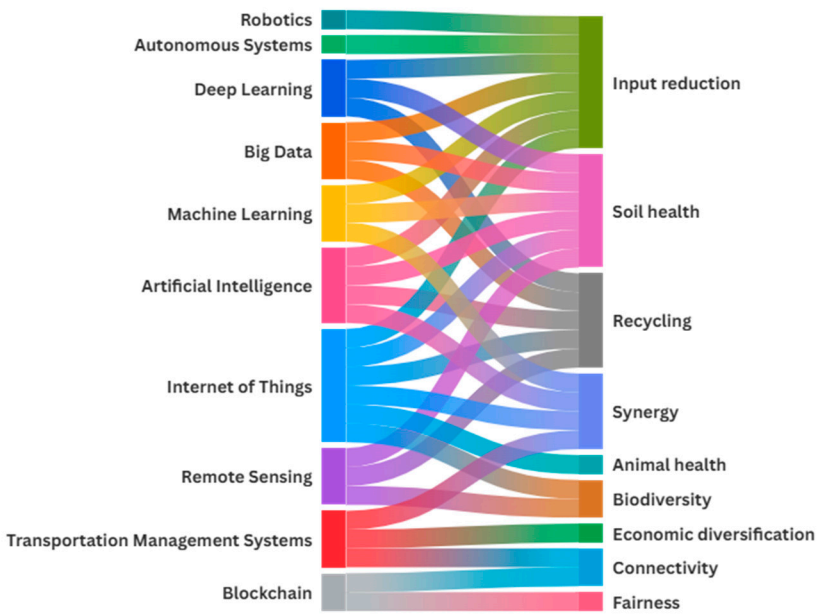

Figure 1. Sankey diagram showing the identified synergies between digital technologies (left) and agroecological principles (right).

There is growing recognition that digital innovations and agroecology can work synergistically to promote sustainable farming practices. Precision agroecology, which integrates

DT with traditional farming knowledge, offers a promising pathway toward more sustainable and resilient food systems [10]. Digital tools play a pivotal role in operationalizing agroecological principles by improving farming efficiency, enabling precision agriculture, and facilitating nutrient cycling at both farm and landscape levels [7]. Recent studies have emphasized the transformative potential of DT in advancing agroecological methods. Tools, such as Transportation Management Systems and the Internet of Things (IoT), are particularly effective in managing logistics and product flow within circular supply chains [11]. Meanwhile, machine learning supports environmentally conscious decision making, helping to optimize resource use, enhance traceability, and reinforce sustainable practices [12]. Additionally, ecosystem services researchers contribute significantly to evaluating the environmental impacts of digital agriculture, informing processes of responsible innovation [13].

It is essential to develop socially equitable technological solutions tailored to the needs of small-scale farmers with limited resources [14]. Ensuring that digitalization supports rather than undermines agroecological principles is critical [15]. Recent research emphasizes the importance of blending traditional knowledge with digital tools to create sustainable and inclusive farming systems [16]. In this context, DA emerges as a promising strategy to address the pressing challenges of sustainable agriculture and food security. By leveraging DT, it is possible to enhance farming efficiency, reduce environmental impacts, and promote ecological balance. However, realizing its full potential requires addressing the financial, technological, and knowledge-related barriers outlined above. This calls for targeted policies, farmer-centered initiatives, and collaborative efforts across sectors.

Conducting research at the intersection of DT and agroecological methods provides valuable insights into sustainable agricultural practices and helps assess the level of technology adoption. This review focuses in a general way on Europe and Latin America, not simply for their geographic diversity but because they embody contrasting yet complementary expressions of agroecological development; furthermore, both regions face a shared structural constraint: the hyper-fragmentation of land. Moreover, these two regions reflect different modes of agroecological evolution, Europe with its emphasis on scientific research and policy frameworks and Latin America with its strong grassroots and movement-led approach centered on food sovereignty and farmer autonomy [17,18].

In addition, Ecuador as an illustrative country case is discussed in greater detail because the characteristics (geographic position, multi-environment sites, biodiversity) of this territory are very suitable for the implementation and study of agroecological practices; however, advancing the adoption of these practices has not proceeded as expected. In Ecuador, 80% of food production comes from peasant familiar agriculture; furthermore, 60% of farms are smaller than 5 hectares [19], a pattern that mirrors the predominance of smallholder farming systems across Latin America and some places in Europe. This structural trait directly affects the adoption of digital tools and agroecological practices, making this country a good case for exploration.

The characteristics and contrasts mentioned above enrich the analytical scope of this review, allowing for an understanding of how digitalization can align or conflict with local agroecological dynamics across distinct political frameworks and reality contexts.

2. Materials and Methods

This study employed a scoping review methodology to explore how DT are integrated into agroecological approaches in Europe and Latin America, with a focus on Ecuador. The review was conducted following the PRISMA-ScR (Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews) guidelines [20] (Supplementary Materials). English and Spanish articles were included using the bibliographic databases

Web of Science, Scopus, and other databases and institutional websites. The search terms used were (1) Agroecology, (2) Agroecological transition, (3) Digital agriculture, (4) Digital agroecology, (5) Digital, (6) Europe, (7) Latin America, (8) Ecuador, (9) Public policy, (10) Adoption, and (11) Polity support (Table 1).

Table 1. Terms applied to the article search of how DT are integrated into agroecological approaches in Europe, Latin America, and Ecuador.

Main Search Terms	1. Agroecology 2. Agroecological transition 3. Digital agriculture 4. Digital agroecology 5. Digital tool	6. Europe 7. Latin America 8. Ecuador 9. Public policy	10. Adoption 11. Policy support
Combination of Terms	(1 OR 2 OR 3 OR 4 OR 5) AND (6 OR 7 OR 8 OR 9 OR) AND (10 OR 11)		

The scoping review process consisted of four stages: identification, screening, eligibility, and analysis (Figure 2).

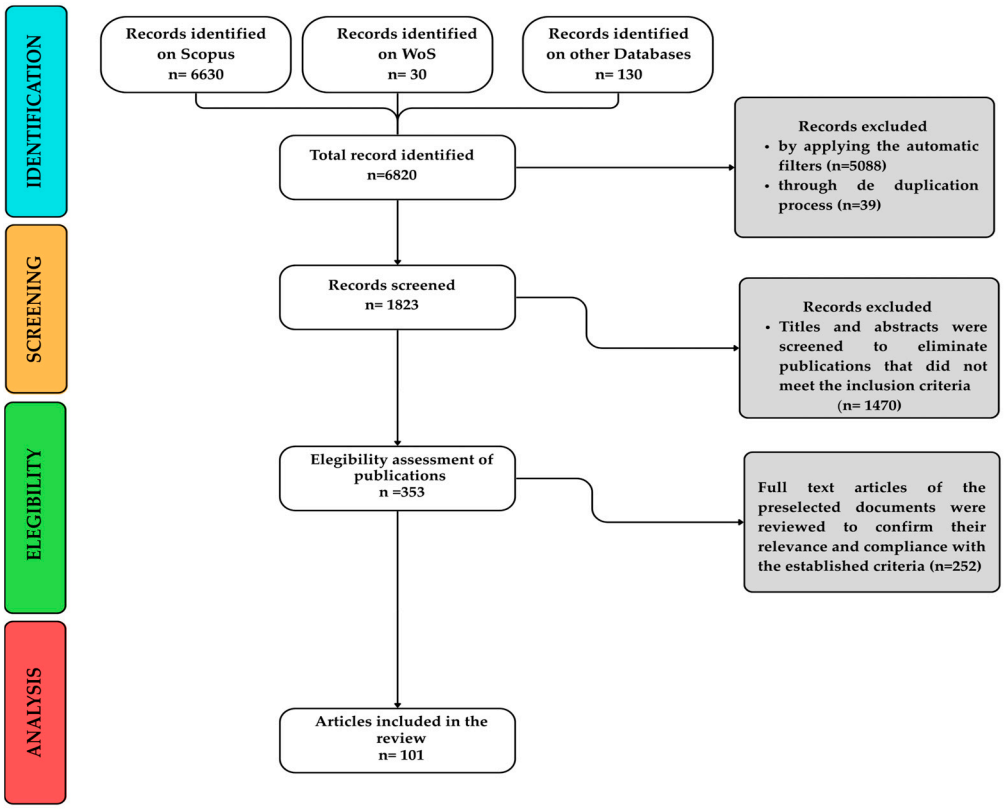

Figure 2. Article selection process using the PRISMA-ScR exploratory methodology.

The search query was based on Boolean operators applied to the following search fields: article title, abstract, and keywords. In the identification stage, a total of 6820 entries were retrieved from all databases (Table 2). The identification stage was further refined by applying automatic filters in each database (Table 3). After filtering, the records were imported into Excel for duplicate removal. Subsequently, inclusion and exclusion criteria were applied to conduct the screening and selection process, as outlined in the PRISMA-ScR flow diagram. Full-text articles of the preselected records were then assessed to confirm their relevance and adherence to the established eligibility criteria.

Table 2. Query formulation for bibliographic databases.

Databases	Search Query	Records
Scopus	(agroecology OR "agroecological transition" OR "digital agroecology" OR "digital agriculture" OR "digital tool") AND (Europe OR "Latin America" OR Ecuador OR "public policy") AND ("adoption" OR "policy support")	6630
Web of Science	TS = (agroecology OR "agroecological transition" OR "digital agroecology" OR "digital agriculture" OR "digital tool") AND TS = (Europe OR "Latin America" OR Ecuador OR "public policy") AND TS = ("adoption" OR "policy support")	30
Other databases	agroecology OR "agroecological transition" OR "digital agroecology" OR "digital agriculture" OR "digital tool"; Europe OR "Latin America" OR Ecuador OR "public policy"; "adoption" OR "policy support"	130
Total		6820

Table 3. Screening benchmarks for initial inclusion.

Databases	Automatic Filters	Records
Scopus	Publication year: from 2010 to 2025 Language: English and Spanish Subject area: Agricultural and Biological Sciences Countries of Europe and Latin America	1565
Web of Science	Publication year: from 2010 to 2025 Language: English and Spanish Area: Agricultural Countries of Europe and Latin America	7
Other bases	N/A	130
Total		1702

During the subsequent screening phase, 353 records were independently reviewed by two authors. This process involved examining titles, abstracts, and keywords using two main criteria: (i) a focus agroecological principles, practices, or transitions, (ii) explored the application of digital technologies in agriculture (e.g., IoT, digital platforms, big data, sensors, AI, etc.); and/or (iii) discussed public policies, governance, or regulatory frameworks related to agroecology and/or digitalization situated in Europe, Latin America, and, specifically, Ecuador regardless of publication year up to 2025. Following full-text assessment in the eligibility phase, a total of 101 publications met the criteria and were included in the final analysis.

3. Relevant Sections

3.1. Digital Agriculture Approach

Digital agriculture, sometimes referred to as the "Fourth Industrial Revolution" or Agriculture 4.0, encompasses a suite of emerging technologies poised to transform agricultural practices in the coming years [21]. These technologies include machine learning, remote sensing, big data, deep learning, robotics, and autonomous systems, many of which are accessible via mobile devices and social media platforms [22]. The ongoing digitization

of agriculture is made possible by the rapid advancement of computing power and global connectivity for data transmission [23].

A key manifestation of this transformation is the rise of "smart farms", which are expanding rapidly worldwide; they aim to reduce labor constraints and make agricultural operations more predictable, modern, safe, and profitable. These farms achieve this by collecting and analyzing large volumes of data from yield monitors and sensors integrated into various stages of agricultural production [8].

A wide range of digital agriculture technologies is being developed for both large-scale industrial farms and smallholder operations [24]. For example, while large farms may use advanced automation panels, small-scale farmers can benefit from simpler technologies, such as fertigation systems. These innovations aim to improve resource efficiency and enhance productivity and profitability across different scales [21]. Additionally, new initiatives (such as digital advisory services offering climate information, market access, and financial tools) are emerging to support farmers through digital means [25].

Importantly, digitalization can align with agroecological farming, but only if it adopts a fundamentally different approach from that used in conventional agriculture [26]. These authors argue that DA should not be a one-size-fits-all solution. Instead, it must embrace diversity, interoperability, sensitivity to local contexts, and adaptability. Smart and precision farming technologies (such as smartphones and tablets providing real-time data on weather, soil, and resource use) can support informed decision making and improve agroecosystem management [23]. Ultimately, digital agricultural tools must harmonize with agroecological principles by recognizing that farmers are not just users but also originators and co-creators of knowledge [26]. This perspective is essential for ensuring that digitalization supports rather than undermines the values of agroecology.

Emerging technologies, such as high-resolution remote sensing and in situ sensors, offer powerful tools for monitoring agroecological systems, which are inherently dynamic. These technologies can track environmental changes and biological processes that influence crop yields, biogeochemical cycles, and water regulation [27]. The use of such high-performance tools generates vast amounts of data, which must be integrated into monitoring frameworks and decision support systems to be effectively utilized [28].

However, the global adoption of these technologies remains uneven. In Latin America, many countries face economic constraints that limit the uptake of agricultural technologies [29]. Even in technologically advanced nations in Asia, adoption rates remain low due to socioeconomic barriers, such as limited access to capital and inadequate infrastructure [30]. Similarly, economic factors continue to pose significant barriers to widespread adoption in European countries [5].

3.2. Opportunities for Digitalization in Agroecology

DT offer a wide range of opportunities to enhance both agricultural and agroecological systems. These include improving decision making, increasing connectivity and data sharing, enhancing productivity and quality, enabling farm monitoring and traceability, reducing labor costs, minimizing environmental impacts, and supporting compliance with standards and regulations [31,32]. While these technologies are rapidly advancing and often associated with the industrialization of agriculture [28], they also hold potential for advancing environmental sustainability, particularly when developed using open-source and open-access models [33]. The use of open-source hardware in agroecology is itself an innovative approach, enabling the development of accessible tools tailored to local needs [34]. Evidence from agroecological transitions shows that farmers adopting these methods often experience increased yields, improved food security, and greater resilience to market and environmental shocks [35]. Advanced digital tools can help manage

environmental variability and reduce the risks and costs associated with agroecological transitions, thereby accelerating their adoption [36].

It is therefore counterproductive to frame agroecology and digital technology as opposing forces. Agroecology is a constructive set of practices, while DT are resources that can be mobilized to achieve sustainable agricultural goals [7]. However, agroecology continues to face challenges, such as weak policy support and institutional resistance. A shift in perception recognizing agroecology as a legitimate and essential component of agricultural policy could significantly enhance its integration into mainstream practices [35].

To bridge the gap between digital innovation and agroecological practice, it is crucial to build strong alliances and collaborative mechanisms. These should support the development of public policies, research agendas, and implementation strategies for digital tools. Effective governance structures involving diverse stakeholders and stages of digital strategy development are essential. Assigning clear responsibilities to existing or new organizations for coordinating regional digitalization efforts will be key to advancing the digital agroecological transition [29].

Agroecological digitalization must consider social and cultural processes related to agriculture rather than supposing that all farm activities are characterized by heavy work and difficulties. It must also recognize and appreciate all dimensions and outcomes that can be provided by participants. This involves including farmers in the design of digital tools that work for them and respecting local cultural frameworks and conditions [8].

The European Commission frequently mentions the pivotal role of the digital transformation of agroecology in rural areas as part of the transformation of food systems, and, to achieve this goal in the digital age, it has highlighted that digital technologies have the potential to revolutionize agriculture by helping farmers work more accurately, efficiently, and sustainably [37]. In this case, digitalization aims to make agricultural jobs more attractive to younger generations and offer consumers greater transparency about how their food is produced. The Common Agricultural Policy (CAP) reform aims to drive this digital revolution through significant investments in agricultural knowledge and innovation systems (AKIS) and farm advisory services (FAS), among other investment areas; in addition, it includes digital obligations, such as the mandatory use of the farm sustainability tool for nutrients (FaST) by income support recipients [23].

On the other hand, in Latin America, DA provides the opportunity to develop open-source or scalable digital tools that break with the economic model of accumulation or the dominance of seed or chemical technology companies, with the aim of generating surpluses by collecting and selling user-compiled data and competing with the tools offered by these latter actors. However, this field has not been developed, and it presents an interesting research alternative for the sustainable food production system [38]. Nevertheless, more training in DT is needed for farmers to achieve this purpose of digitalizing practices to generate agroecological outputs [8]. Additionally, in Latin America, despite the potential of digitalization, the lack of adequate infrastructure in rural areas and high implementation costs exclude many smallholder farmers; only 45% of rural households have access to the internet, perpetuating inequalities and affecting the autonomy of local communities [39].

3.3. Case Studies on Digital Agriculture Adoption

3.3.1. Agroecology Approach in Europe

Agroecology as a scientific discipline appeared in the 1930s in Europe; however, it was not before the 1970s that critical people became interested in the ecological effects of agriculture [40]. According to the study by Gallardo-López [17], it is considered a science in the Netherlands, Portugal, and Romania; it is conceived mainly as a science, followed by practice and with marginal conception as a movement, in Spain, Switzerland,

Denmark, Germany, Poland, Austria, and the United Kingdom (UK); and it is considered a predominant practice, followed by a science and, to a lesser degree, a movement in Ireland, Lithuania, Belgium, the Czech Republic, Norway, Finland, Slovakia, Iceland, Hungary, France, Sweden, and Italy. Therefore, research and education are major components of agroecology in Europe, but, often, there is a bridge between its manifestations of science and practice [41].

There are relevant institutes and universities in Europe developing agroecological research and technology [41]. For example, Agricultural Research for Development—CIRAD (France) deals with agroecology in the tropic and subtropic areas; Wageningen University (Netherlands) has the Centre for Agroecology and Systems Analysis to research global food security, animal production, natural resources, bio-based economies, and climate change; Coventry University (UK) has the Center for Agroecology, Water and Resilience, which drives innovative and transdisciplinary research; and the Belgian National Scientific Research Foundation encourages food production and the development of knowledge. There are more institutes and universities around Europe designated to research agroecology-related topics, such as soil, water, ecosystems services, farming practices, agroecosystems, and food systems, and develop new technologies like digitalization for food production.

DT have the potential to help farmers operate more precisely, efficiently, and sustainably, thereby contributing to the modernization of agriculture [37]. The European Union (EU) envisions digitalization as a means to make agriculture more appealing to younger generations and to provide consumers with greater transparency regarding food production [23]. Digital tools and precision equipment are particularly valuable for monitoring agroecosystem dynamics. They enable modeling of cropping practices and the spatial and temporal dimensions of ecosystem services [28]. Despite this potential, there is a notable lack of data on agroecological systems and transition processes in industrialized countries. Digital agriculture, remote sensing, farm networks, and spatio-temporal data analysis can help address this gap by supporting experimental approaches focused on agroecological principles [42]. In the European context, agroecological practices are often associated with organic farming [23], as in the other regions of the world.

In the UK, agroecology is increasingly recognized as a key component of climate-smart agriculture and sustainable intensification. It is valued for its ability to integrate environmentally friendly practices with mainstream industrial farming technologies [43]. This includes agroforestry, intercropping, and other practices, alongside the use of herbicide-tolerant crop varieties, low-toxicity pesticides, genetic engineering, proprietary technologies, energy-intensive systems, digital farming, carbon rights, and biofuel plantations [43,44]. In Spain, the CONECT-e digital platform represents an innovative approach to preserving and sharing traditional ecological knowledge. By involving farmers directly and operating under a digital commons framework, the platform ensures open access to knowledge and prevents its privatization, thus supporting the agroecological transition [45]. Several European countries, including Italy, Estonia, Poland, and Romania, have implemented agroecology living labs through initiatives like the ALL-Organic Project. These labs test crop diversification strategies in both experimental and real-world farm settings, enabling the study of functional socio-technical interactions [46].

A notable international initiative is the H2020 MSCA-RISE UNDERTREES project "Creating knowledge for Understanding ecosystem services of agroforestry systems through a holistic methodological framework", which aims to build a global research network focused on agroforestry and ecosystem services. This project includes partners from Europe (UK, Italy, Spain), Africa (Ghana, Tanzania), and South America (Ecuador, Chile) and promotes the use of both conventional technology and DT to support agroecological transitions [47].

By 2025, it is projected that farms in industrialized countries using smart farming technologies will generate millions of data points daily [48]. These data streams (combined with robotics, weather forecasting, and market information systems) enable high levels of automation in farm management [8]. Compatibility between agroecology and digital innovation is real but must be considered carefully [23]. This compatibility depends heavily on who controls the digital tools and for what purposes they are used. The integration of DT into agroecological systems must be assessed on a case-by-case basis to ensure alignment with ecological principles [3] and the specific needs of each agroecosystem [23].

3.3.2. Agroecology Approach in Latin America

Latin American countries are particularly well-positioned for the emergence and development of alternative agricultural models that address social, environmental, economic, and public concerns [49]. Agroecology in Latin America emerged in the 1970s as a response to growing environmental challenges, advocating for a significant transformation of conventional production systems [18]. It promotes a holistic vision that integrates social, environmental, economic, and cultural dimensions to foster a new model of sustainable rural development [50], in contrast to the industrial model rooted in Green Revolution principles [51]. In terms of history [52], the first classes on agroecology were taught in Mexico and Colombia in 1970. Some nonprofit organizations (NGO) with agroecological approaches, such as the Latin American Consortium on Agroecology and Development (CLADES) and the Agroecological Movement of Latin America and the Caribbean (MAELA), were created between 1980 and 1990. The first research about this subject was also developed during this time in Cuba (1993) and Mexico (1998); for instance, the first studies were carried out by the Cuban Association of Organic Agriculture (ACAO), the Cuban Association of Agricultural and Forestry Technicians (ACTAF), and the National Autonomous University of Mexico. In 2007, there was a consolidation of the academic field of agroecology through the creation of the Latin American Scientific Society of Agroecology (SOCLA). Several policies, norms, and laws for developing agroecology and organic farming have been created since 2001 in Latin American countries [50].

Key research areas in Latin America include agroecological management, biodiversity conservation, local self-management and self-sufficiency, soil and water conservation, and agroecological education. Less emphasis is placed on forest cover conservation, carbon sequestration, and agricultural equity [53–55]. However, the extent to which digital tools are integrated into agroecological research in the region remains unclear. Brazil leads in agroecology research, followed by Mexico, Colombia, Argentina, Cuba, Ecuador, and Chile, while countries like Guatemala, Honduras, and Paraguay show lower levels of research activity [53].

While in Europe agroecological research approaches led to the publication of scientific papers and books and the implementation of education programs, in Latin America, it started with hands in the soil, long before the traditional management of agroecosystems received the stamp of agroecology [40]. Therefore, scientists were defining concepts and classifying practices in Europe, and the scientific component of agroecology remains the most recognized, whereas Latin America was debating autonomy and food sovereignty in the field, where practice and movements have historically been the leading forces and are still mostly dominant [41].

Latin American experiences can inspire agroecological initiatives in other regions, such as Africa and Asia, by encouraging the restructuring of intervention strategies to support agroecological transitions [50]. Several innovations have emerged to enhance agroecological performance based on sustainable production, community organization, gender equity, digital technologies, and the integration of traditional knowledge with

scientific research [56]. For example, Colombia's Lab Campesino initiative strengthens local innovation ecosystems and promotes agroecological practices. It includes a free hardware digital kit for monitoring humidity and temperature in biofertilizer production, helping to optimize bio-input development [34]. Experiences, such as the IPA-LAC project, seek to promote agroecology through participatory research in the region, while the SANE project supports the agricultural development process in rural sectors through institutional coordination and the training of farmers and technicians as a strategy to promote agroecological principles in countries like Cuba, Peru, and Chile [57].

Digital platforms for e-commerce are also gaining traction. In Brazil and Chile, such platforms are helping to establish direct connections between farmers and consumers, including digital marketplaces for land reform settlements [58]. However, adoption is often limited by farmers' financial constraints and low levels of digital literacy [59]. In Argentina, state policies are being developed to create, adapt, and validate technologies tailored to the needs of family farming. These efforts contribute to a more inclusive and sustainable agroecological transition [60].

Scientific interest in digitalization and agroecology in general has increased in recent years in Latin America because it is emerging as a way to integrate agroecological principles with digital technologies to transform food systems [38]. This combination seeks to improve the efficiency, sustainability, and resilience of agriculture while promoting social justice and food security. However, it remains unclear if DA as a solution for scaling agroecology can significantly shift the power dynamics within the food system by creating new avenues through which farmer organizations can design and adapt locally relevant solutions to the challenges facing their own food systems while also building on culturally specific indicators and certification frameworks [8]. Digitalization is considered one essential pillar of future agricultural developments in Latin America [38]; for this reason, it is necessary to improve digital literacy and capacity across all demographic dimensions, especially for marginalized populations [8]. Currently, agroecology movements in Latin America are already using digital networks to communicate and exchange experiences [38]; however, more training and dissemination of DT is required to make possible a successful transition.

Consequently, countries with emerging economies can harness the potential of technological adoption to generate positive social impacts by embracing collaborative strategies. However, this requires a comprehensive approach that integrates research, practical applications, and policy development, fostering a digitally empowered and more inclusive agricultural sector [29].

3.3.3. The Case of Ecuador, a Country with Agroecological Potential

Ecuador, the smallest country in South America and part of the Andean region, is recognized for its megadiversity in flora and fauna. Agriculture plays a vital role in its economy, and the implementation of agroecological principles is essential for conserving its rich natural resources. In the Andean highlands, agroecological systems have demonstrated significant improvements in soil quality, including increased organic matter and water retention, reduced erosion, and enhanced long-term fertility [61]; in addition, social movements have played a key role in advancing agroecology in Ecuador [62].

Agroecology has proven to be an effective strategy for promoting environmental sustainability in Ecuador by mitigating the negative impacts of conventional agriculture [61]. Although agroecological food production and consumption are growing (supported by local market distribution systems), they remain marginal within the broader food supply chain [63].

From the 1980s onwards, agroecological proposals, usually labeled as "alternative agriculture", started in Ecuador. The first organization that began to implement biological

and organic farming was the Ecuadorian Corporation of Biological Producers (PROBIO), and the definition of the agroecological technological principles allowed its integration with ancestral agriculture [62]. In the 1990s, agroecology incorporated social and political dimensions, and the necessity of an articulation at the national level was recognized to link it to the development of peasant models in a way that superseded mere technological support. From 1996 to 2006, agroecology was more broadly disseminated, and it obtained a more comprehensive and multi-dimensional focus based on developing training programs for farmers, technicians, and agroecological promoters. The academic sector was more enthusiastic with this approach, and the first National Agroecology Summit was carried out in Quito in 2006, where a new phase started in Ecuadorian agroecology history with the articulation of new actors. From 2006 to 2016, agroecology had a new emphasis on questioning the notion of progress and the supposed neutrality of science and technology. The Agroecological Collective of Ecuador (CAE), formed in 2007, promoted more political connections between city and countryside, as well as the development of policies related to this approach. Therefore, the agroecology approach was included in the National Constitution in 2008 [62]. Since 2009, the Food Sovereignty System Comprehensive Law (LORSA) has provided a legal framework for supporting agroecological farmers through training, resources, and market access [64]. More recently, the Organic Law on Agrobiodiversity, Seeds, and Agroecological Promotion (LOASFAS) has taken initial steps toward regulating open-access knowledge related to agroecology [65].

Currently, the UNDERTREES project and Ecuadorian research funding have supported several studies by the National Institute of Agricultural Research (INIAP) on agroecology and agroforestry systems. These studies focus on economically important crops, such as cocoa, coffee, pitahaya, and naranjilla, in the Amazon region [66–70]. The research highlights the benefits of agroecological practices in producing safe food, protecting the environment, and enhancing carbon sequestration. Importantly, findings show that well-managed agroecological farms can achieve income levels comparable to conventional systems while delivering superior environmental outcomes.

There have also been initiatives in Ecuador to develop digital platforms for managing agroecological knowledge and fostering online communities. These platforms follow international standards for agricultural information management and aim to provide free and universal access to digital content (metadata), thereby supporting research and education [63]. E-commerce has also emerged as a tool for marketing agroecological products. For example, the commercial brand GranSol involves 183 Ecuadorian agroecological farmers, while the Union of Agroecological Producers and Associative Commercialization of Tungurahua (PACAT) includes 26 associations and 420 families [63]. These groups use digital tools and networks to distribute their products nationwide. Interestingly, the prices of agroecological products are often comparable to, slightly higher than, or even lower than those of conventional goods [71,72]. Despite these advances, significant barriers remain. A study by Cusme and Gaibor [73] found that while 60% of surveyed farmers expressed moderate interest in agroecology, over 80% reported lacking sufficient information to make the transition. This indicates that there is a significant gap between interest in and practical implementation of agroecology among farmers because there is little government support due to the lack of public policies that promote the sustainable development of family farming through the implementation of agroecological practices, the lack of organization and active participation of the rural community in decision making, low levels of training, and limited access to resources and technologies; thus, greater opportunities are needed to strengthen the participation and commitment of the community in strengthening agroecological farming and its impact on sustainable rural development. The lack of information and training is a major barrier to the adoption of agroecology, especially in rural areas.

Farmers need access to technical knowledge, information on the benefits of agroecology, and advice on how to implement these practices on their own farms. Training and technical assistance programs, as well as the dissemination of successful experiences, can play a crucial role in closing this gap and encouraging greater adoption of agroecology in Ecuador. Consequently, although there is interest in agroecology among farmers, investments in resources and efforts are needed to provide farmers with the knowledge and support needed to transition to more sustainable agricultural systems.

Key challenges include limited funding, cultural resistance, lack of digital literacy, resistance to change, insufficient incentives, and the absence of a comprehensive public policy framework. To align with Ecuador's constitutional goals of food sovereignty and good living, public policies must be strengthened to scale up agroecological practices. These policies should aim to improve smallholder incomes while ensuring environmental and economic sustainability [68]. Currently, Ecuadorian Universities, such as the Amazon Regional University (IKIAM), the University of Azuay, and Quevedo State Technical University, among others, offer technology courses and undergraduate and master's degrees focused on agroecology, which means that this approach is being positioned in the country's academy.

Access to DT remains limited, particularly in some rural areas, which hampers the dissemination of agroecological knowledge [74]. However, there are promising opportunities, including crop diversification, access to organic markets, and the growth of rural tourism. With the right support policies and training programs, agroecology can serve as a viable strategy for enhancing sustainability and improving quality of life in rural communities [61].

3.4. Policy for an Agroecology Approach

Policy frameworks play a pivotal role in scaling DA. Although some European policies support agroecological practices, such as agroforestry and mixed farming, large-scale conventional agriculture persists in several sites, and direct support for DA remains limited [75]. In Latin America, agroecological policies are still underdeveloped, and extensive conventional agriculture continues to dominate [50]. Key policy constraints include insufficient research and extension services, lack of incentives (such as payments for ecosystem services), insecure land tenure, and agricultural and trade policies that favor the agro-food industry [35].

The EU's CAP supports the digital transition through significant investments in agricultural knowledge, innovation systems, and advisory services. It also mandates the use of digital tools, such as the "Farm Sustainability Tool for Nutrients", accessible via mobile devices and computers [23]. European policy frameworks are well-positioned to incorporate inclusivity and ethical considerations—key elements for integrating digitalization with agroecological transitions [76]. A comprehensive community-based framework is essential, especially one that considers natural, cultural, human, and social capital. This approach ensures that data generated by farmers and other stakeholders are meaningfully connected to agroecological principles across all components of the production system (agricultural, livestock, agroforestry, conventional, or organic) [77]. The European policy is addressed to improving the adoption of agroecological practices in the region through initiatives like the "European Green Deal" and funding various projects dedicated to promoting agroecological research within the framework of the Horizon 2020 program [78]. Although this policy aims for the promotion of agroecology, the interventions are often too low or unattractive to farmers and sometimes even conflict with sustainability objectives. Thus, agroecology must be integrated into rural development, environmental, health, and trade policies. It is crucial to increase economic incentives for agroecological practices, such as

payments for ecosystem services, and ensure that farmers have access to financing for the transition. Furthermore, it is necessary to invest more in the research and development of agroecological technologies, as well as in training farmers and technicians in these practices.

In Latin America, several countries have implemented public policies, norms, and laws to promote agroecology [52]; some of the countries with notable initiatives are Argentina (Law for the Promotion of Agroecological Production), Brazil (National Plan for Agroecology and Organic Production), Chile (National Law on Organic Production), Paraguay (National Strategy for the Promotion of Organic and Agroecological Production), Costa Rica (Organic Agriculture Law No. 8591 in Costa Rica), Cuba (Food Sovereignty and Food and Nutritional Security Law N° 148/2022), El Salvador (Family Farming Law Decree No. 814), Mexico (Law to manage organic agriculture), and Nicaragua (Law for the Promotion of Agroecological or Organic Production). In Peru, there are regulations (Law No. 29196, Supreme Decree No. 044-2006-AG, and Supreme Decree No. 002-2020-MINAGRI) that promote the sustainable and competitive development of organic agriculture, improving the quality of life of farming families and reducing rural poverty. Ecuador also has this kind of policy included in its Constitutional Law [62]. This transformation calls for a shift from conventional to sustainable systems; however, while some policies support organic and sustainable agriculture, they often fail to challenge the foundations of conventional systems. As a result, these policies remain fragile, with conventional agriculture continuing to receive the majority of institutional support [50]. Consequently, strengthening policy frameworks is essential to facilitate and accelerate the agroecological transition [52]. In addition, it is crucial to involve farmers, consumers, and other stakeholders in policy design and implementation to ensure their success.

Although these policies remain fragile, they seek to foster sustainable agricultural production, diversify food production, and strengthen family farming; in addition, currently, there is continuous acceptance of agroecological practices among farmers [79], but more work and research are needed to improve the implementation of agroecological principles and the digitalization of this approach.

4. Discussion

4.1. Limitations, Challenges, and Barriers to the Adoption of Technologies

Overall, agroecology has challenges, such as the dominance of industrial agriculture, market pressures, and deficient policy support [1]. The agroecological transition is a complex and multifaceted process that requires the alignment of technological innovation with agroecological principles [80]. It remains unclear if digital agriculture for scaling agroecology can considerably change power dynamics within the productive systems, as this co-evolutionary process involves navigating tensions between DT (often associated with industrial agriculture) and the values of agroecology, such as transparency, equity, participation, and the co-construction of knowledge [8]. For this reason, state involvement is essential in creating and implementing public policies that facilitate the adoption of agroecological technologies [81].

The widespread adoption of DA faces several limitations, challenges, and barriers (Table 4). Financial constraints and economic feasibility pose significant barriers for farmers and stakeholders, especially in resource-limited regions [82]. Moreover, attitudes toward the economic performance of DT influence adoption rates. However, factors like innovative behavior, proactive information seeking, and supportive policies—like subsidies and tax incentives—can serve as positive drivers of adoption within farming communities [5].

In the different regions of the world, the main factors leading to low adoption of DT are the lack of financial resources, absence of subsidies, and inadequate infrastructure [5,30], as well as technological and knowledge barriers, as comprehensive understanding of

digital tools and their applications make difficult a fast transition from the conventional approach to the DA [83]. Furthermore, data reporting without agronomic interpretation is not meaningful or useful for farmers because they do not have the time or training to carry out data analysis; therefore, interpretation of the data is required to translate the use of digital tools and technologies into confident, efficient, and relevant action [48].

Agribusiness companies have developed software capable of data interpretation for farms to face these concerns, providing data-driven management analytics and solutions for producers and trying to fill the current gap in public extension and agronomic advisory services worldwide [8]. However, existing incompatibilities between digitalization and agroecology are highly associated with the substantial role of huge private sector actors in developing technological solutions [23].

Cybersecurity has to be considered to protect computer and informatic systems, networks, programs, and data from digital attacks, damage, or unauthorized access. Cybersecurity threats are increasingly relevant as digital farming systems become more reliant on cloud-based platforms, sensor networks, and data-driven decision making tools. These systems are vulnerable to data breaches, manipulation, or service disruption, which can compromise not only farm productivity but also farmer trust in technological solutions [21,31]. Nowadays, innovative projects incorporating agro-cybernetics are been developed to create an eco-intelligent network for resilient agroecology where data flow effortlessly between the physical and digital realms, with the objective of revolutionizing traditional farming practices by integrating cybernetics principles with advanced IoT technology [84]. The framework looks to empower farmers with data-driven decision support, promoting responsible resource management and encouraging agricultural practices that coexist coherently with the environment. In this context, massive data collection and processing generate ethical questions about privacy, information security, safeguarding data, and having control over shared information [85]. Regarding data use, family farmers could fear or experience difficulties maintaining ownership and control of their data, which can lead to privacy issues, such as the exploitation of sensitive information [86]. Therefore, controlling these data by large monopolies disadvantages small producers and smallholder farmers. Consequently, the transition to digital agriculture shows potential benefits, like the optimization of resource use and enhancing efficiency; however, it also triggers deep concerns related to data privacy, unequal access to digital tools, technology dependency, and the increasing concentration of power in agribusiness and large technology corporations. These processes risk aggravating the vulnerability of smallholder farmers, enlarging socioeconomic inequalities, and decreasing their autonomy in agricultural production [85].

Perceptions of technological complexity also act as a barrier, particularly in the traditionally conservative agro-food sector [87]. Technological and knowledge-related barriers also pose significant challenges to the adoption of DA. Many farmers lack a comprehensive understanding of digital tools and their practical applications, which slows the transition to digitally supported agroecological systems [83]. The absence of technical guidance and support further complicates adoption efforts. Addressing these obstacles requires tackling issues, such as limited awareness, inadequate resources, and policy shortcomings [1]. While some proponents view digital farming as a key strategy for promoting environmental sustainability, the reality is more nuanced. As Miles [88] suggests, precision agriculture, often seen as a cornerstone of digital farming, may represent more of an "evolution" than a "revolution" in agricultural practices. Given the potential drawbacks and the broader socioeconomic context of agricultural production, it is essential to critically assess the environmental benefits of digital farming.

Additional challenges include increased energy consumption, inconsistent adoption rates across stakeholders, lack of interoperability, and weak data protection frameworks [89].

Moreover, many farmers lack the organizational capacity and experience needed to effectively implement digital tools, which further complicates the transition [90]. There are few sustainability characteristics incorporated into accessible digital technical advisories for smallholders in low- and middle-income countries, and major knowledge gaps are predominant. Tools do not take account of progressive agriculture content, and there is limited robust scientific evidence behind recommendations; therefore, a better effort to include diverse and novel practices for agroecology, including topics like climate change adaptation and mitigation, is required to strengthen the resilience of smallholders [76].

Geopolitical risks likewise pose systemic challenges; disruptions in global supply chains, trade sanctions, and political instability can limit access to hardware, digital infrastructure, or even data sovereignty, particularly in countries with emerging economies [32]. The geopolitical position of Latin America due to its contrasting features characterize the region because it has the planet's largest tropical forest and the greatest amount of space for agricultural expansion, with potential agroecological impacts [91]. Food sovereignty is a policy pillar in countries like Colombia, Peru, Ecuador, and Bolivia; however, actual policy implementation privileges national-scale self-sufficiency over the protection of local and sustainable agriculture and territorial rights [91], influencing directly the adoption of agroecological principles. On the other hand, geopolitical risks and environmental policy have become increasingly important in the EU, which is committed to tackling climate change and protecting the environment through the implementation of agroecological practices; however, geopolitical risks can undermine its environmental policy objectives. They notably influence environmental policy in countries like Denmark, Estonia, Finland, France, Germany, Luxembourg, and Romania; therefore, policymakers must develop resilience to geopolitical risks by promoting agroecological outputs, such as renewable energy, sustainable resource use, environmental conservation, food safety, and social and economic development [92].

Concerns persist regarding data sovereignty and the need to prioritize farmers' knowledge and autonomy in the digital transition [8]. Persistent gaps and ongoing debates about the role of digital agriculture in agroecology highlight the need for further research and dialogue [33]. It is essential to better understand how DT contribute to agroecosystem improvement and to assess the extent of their adoption by farmers. If digital tools can indeed support agroecological transitions and generate public goods, then their integration into food systems becomes not only viable but necessary [23].

In addition to the commonly cited limitations, such as financial constraints, low digital literacy, and insufficient technical support, the adoption of digital technologies in agroecology must also contend with broader enterprise risks that remain underexplored in the current literature. Moreover, health and safety constraints, especially those arising from prolonged human–machine interaction, increased electromagnetic exposure, or unsafe equipment use, must be addressed to ensure that digital tools are designed and implemented with occupational safety in mind. Recognizing these systemic risks is essential for designing robust and inclusive agroecological digitalization strategies that safeguard not only ecological and economic outcomes but also institutional and personal resilience.

Table 4. Limitations, challenges, and barriers to the adoption of DA: Ecuador, Latin America, and Europe.

Category	Ecuador	Latin America	Europe
Economic Barriers	Limited credit access for smallholders, high cost of technologies [19]. Lack of subsidies for agroecological practices.	Economic constraints (government and farmers) limit the adoption of agricultural technologies [29,49].	Financial support exists, but uneven; subsidies favor conventional systems [5].
Technical Knowledge	Low digital literacy; limited training and extension services [61,65,73].	Limited adoption of technologies due to lack of training for extensionist services in agroecological matters [74].	Higher education levels, but slow adoption in traditional farming sectors, low digital literacy, and inadequate infrastructure, which can hinder their effectiveness in real-world farming conditions [31].
Infrastructure Gaps	Lack of access to collection centers, quality transportation routes in rural areas, and irrigation systems.	Poor rural connectivity, lack of adequate infrastructure in rural areas, and scarce digital platforms adapted to small-scale farming [21,39].	Infrastructure available, but challenges in integrating across diverse regions [78].
Cybersecurity Risks	Lack of information security, limits to safeguarding data, and overshared information.	Limited awareness and regulatory frameworks for digital safety. Ownership and control of the data. Exploitation of sensitive information [21,85].	Emerging concern due to increased data reliance and interoperability needs [31].
Geopolitical Risks	Political instability, petroleum dependence, regulatory uncertainty, volatility in international markets, and uncertainty in rural land ownership.	Vulnerable to global supply chain disruptions; dependence on foreign tech; policy implementation privileges national-scale self-sufficiency [32,91].	Trade conflicts and regulatory environmental policy differences affect tool access and data governance. Geopolitical risks can affect environmental polices [92].
Health and Safety Constraints	Farmers are not familiar with the risks of using sensors, drones, or digital platforms incorrectly. There are no clear protocols or safety-focused training programs.	Lack of occupational safety protocols in tech usage; unfamiliarity with devices.	Overdependence on automation in safety-critical tasks.
Policy and Institutional Support	Agroecology recognized in law but with weak implementation and incentives [62].	Big efforts to implement policies for agroecology, but they remain fragile [50,52].	Stronger frameworks in place (e.g., CAP), though they often favor large-scale farming [23,76,78].
Farmer Autonomy and Data Rights	Lack of awareness about data ownership, No specific laws protecting agricultural data.	Low capacity to negotiate fair tech use; dependence on third-party platforms [8,85].	Debates ongoing about data sovereignty and private sector dominance [23].

4.2. Benefits of Technologies for the Agroecology Approach

Effective agricultural development requires an agroecological approach that extends beyond crop yields to consider the complex, interrelated factors that contribute to sustainable agroecosystems. This necessitates the development of empirical and conceptual

indicators within a multidimensional and systematic methodology [93]. Technologies play a fundamental role in agroecological innovation systems, not as superficial add-ons but as integral tools for strengthening transition processes [81]. The integration of DT in agroecological management has several benefits (Table 5).

Table 5. DT benefits in agroecology by category and contribution.

Category	Benefit	Contribution
Productive	Cost reduction Precision agriculture Smart irrigation Operational optimization	Lower input use Better yield and quality Monitoring with GPS and sensors Data-driven technical decisions
Environmental	Reduced soil compaction Improved plant nutrition Waste reduction Environmental conservation	Use of GPS-guided tractors Spatial predictive models More sustainable decisions Lower input impact
Social	Smallholder inclusion Consumer connectivity Inequality reduction Rural support	Accessible technologies Digital platforms Promotion of equity E-commerce and digital governance

The digitization of ecosystems has increasing influence over how food is produced, what food people buy, and flows of information among supply chain actors and consumers. Efforts to transform food systems towards sustainability and resilience are progressively dependent on digital resources, and alternative agriculture and agroecology are promoted as a solution [76]. Digital tools offer numerous benefits for food sovereignty and farm networks. For instance, digital land administration systems can enhance tenure security, while digital marketplaces can connect producers directly with consumers, reducing dependency on intermediaries [8]. These technologies also enable the development and dissemination of novel production methods, many of which are accessible and replicable by small-scale producers [81]. Digital agriculture can reduce input costs, improve irrigation efficiency, and boost productivity. E-commerce platforms are increasingly used to open new markets and marketing strategies for agricultural products [8]. Moreover, the adoption of DT can help reduce socioeconomic inequalities by fostering social inclusion and supporting marginalized farming communities [60,94].

Some examples of the implementation of DT for improving agroecological practices are the use of global positioning systems (GPS) in agricultural machines (i.e., tractors), which allows for the reduction of soil compaction (soil stress), which is often detrimental to crop production because it can inhibit crop root growth and reduce the ability of roots to uptake water [95]; in addition, this kind of technology produces other benefits, such as machining uniformity, increased work quality, reduced resource use, and reduced environmental burdens, supporting sustainable agriculture [96]. Spatial yield monitoring can be used to guide future fertilize applications because this technology offers predictive models, adaptive management strategies, and real-time decision support systems, which reduce nutrient runoff, enhance plant nutrition, and improve plant health [97]. In this context, drones are used for crop yield monitoring, nutrient and water stress assessment, weed distribution mapping, and pest management, among others, contributing to promoting the Agriculture 4.0 framework within the different areas of precision agriculture, smart farming, and climate-resilient farming systems [98]. Global information systems (GIS) and remote sensing technologies can offer support for improving agricultural sustainability and identifying obstacles to their application, particularly in low- and middle-income countries. They offer better methods for the analysis of spatial factors that affect agricultural produc-

tion compared to approaches where geographically explicit data are absent; consequently, they can be integrated to enhance agriculture decisions and policymaking [99].

While integrating DT into agroecology is not without challenges, it presents a real opportunity to enhance both organizational and technical capacities. Tools for data sharing and knowledge exchange can significantly improve outcomes in agroecosystems [7]. The creation and implementation of innovative methods and practices can increase productivity, reduce waste, and enhance resilience to climate change, thereby advancing agroecological development through digital means [81].

DT also support the monitoring and management of agroecosystems by enabling real-time tracking of environmental dynamics and resource flows [28]. When developed within an agroecological paradigm, these tools can promote equitable skill building, cooperative learning, resource sharing, and mentorship, practices that often conflict with the objectives of industrial food systems [23].

4.3. The Case of Ecuador

Ecuador's diversity and peculiar characteristics mean that the agroecology approach adopted in different locations differs significantly, making it complex to standardize all processes relating to agroecology. There is a generalized consensus among farmers and consumers that it is a good alternative for defeating current and future food crises [62], and farmers' adoption of such agroecological practices has had a moderate impact on overcoming conventional management [79], but the use of digital technologies for this approach is limited. Agroecology in Ecuador has grown from its early stages as a little-known, unclear reaction to the Green Revolution to a serious agrarian model with fundamental principles included in the country's Constitution and Food Sovereignty Law since 2008 [62]. However, the scope and implementation of agroecological practices framed within these policies have varied because ongoing systems for disseminating and teaching about these principles have not been developed throughout rural areas; furthermore, the limited government personnel in the agricultural sector hinders widespread dissemination of this approach. Consequently, a successful transition to digital agroecology is complex and slow-moving. New agroecological agrarian reforms must be implemented that take into account not only land tenure and sustainable management but also economic incentives, agricultural credit, process autonomy, novel technologies, training and ongoing monitoring of producers, and the strengthening of strategic alliances.

Academia has played an essential role in enabling the introduction of conventional agriculture practices into Ecuador because its curricula content was created to respond to the interests of multinational corporations, training professionals with a limited and simplified vision of agriculture [62]; however, this has changed in recent times. Currently, several Ecuadorian universities offer educational programs that include agroecological content. Although these elements are not weighted the same as those dedicated to conventional agriculture, they have a positive impact in academic circles.

Digital agroecological technology subjects are limited in the formation of new professionals and extensionists, which complicate their diffusion to farmers. In addition, a large percentage of Ecuadorian farmers are between 46 and 75 years old, with a considerable number of people over 76 years [62], which also impacts the learning and adoption of these kinds of new and novel technologies by producers.

Beyond environmental benefits, agroecology has had a positive impact on the social well-being of Ecuadorian rural communities [100]. It has improved food security and access to healthier diets while also reducing production costs through practices like organic fertilization and crop rotation [100]. These savings enable farmers to invest in technologies like irrigation systems and precision agriculture tools that enhance agroecological produc-

tivity [101]. However, only 25% of Ecuadorian farmers have access to agricultural credit, limiting their ability to invest in sustainable technologies [102].

In addition, the country faces several challenges related to agroecology that can be resolved, such as climate change, rural poverty, the need to recover the strength and vitality of agroecosystems, the production of healthy food, and the conservation of agrobiodiversity [62].

4.4. Limitations of This Study

A scientific and technical literature search was conducted using keywords related to the topic; however, there may be information related to the topic that was not considered in the search. In addition, there is a lack of research studies that show specific results (data generated) regarding the advancement or progression of the adoption of DA in countries in Europe and Latin America. In the case of Ecuador, a few studies were found about the theoretical conception of farmers about the implementation of agricultural practices, but there were no data about the adoption of DA in agricultural productive systems taking into account production variables and socioecological aspects, such as farmers' livelihoods or well-being. DA needs to be more closely and directly aligned with agricultural practices; thus, it is necessary to evaluate the technical, social, economic, cultural, and political dimensions. It is also important to carry out economic analyses of the implementation of DT to know their influence on the production costs and modeling scenarios to determine real economic efficiency, including unpaid family labor as a key element to understand the functioning of the peasant or indigenous economy in Ecuador; therefore, further research is required in the aspects mentioned above because despite their potential, the adoption of DT in agroecology remains limited in many countries worldwide.

5. Conclusions

The agroecology approach is positioned globally; however, its implementation has been adopted mainly using conventional methods that must be updated with modern DT that enable the transition to advanced agroecology concepts. Clear, logical, and progressive ideas with a well-organized structure are needed to effectively communicate the benefits and limitations of DA to promote sustainable agriculture. Furthermore, novel strategies must be developed to overcome the aforementioned barriers to the adoption of digitalization, especially by small- and medium-sized producers. Innovative agroecology tools (DT and agriculture precision equipment) are needed for improving farms as well as agricultural production systems (agrosystems); however, this will require a major and better commitment from agronomic research and articulation with concerned actors of agri-food chains while considering aspects related to the medium- and long-term adoption process for this approach, as well as improved training for farmers to obtain knowledge related to new technologies.

Industrialized countries like those in Europe and emerging economy countries like those in Latin America have implemented different forms of action (tools and policies) to support the implementation of an advanced agroecology approach in these regions; however, the adoption level is not enough, and there are financial barriers that stop the transition model. In the specific case of Ecuador, the implementation and adoption of digital tools and technology toward agroecological transition are limited, primarily due to economic constraints and the limited training of small- and medium-sized farmers in using digital tools or equipment. For this reason, greater government intervention is required, with policies and actions that lead to a strong positioning of agroecology and its modern applications to facilitate the transition toward advanced agroecology in terms of digitalization as a tool for improvement.

It is recommended to create spaces for discussion of the benefits and consequences of adopting agroecological practices through DT at different scales, taking into consideration the whole picture, from farms to the entire food-chain system. This will allow for an analysis of the effects on all dimensions (human, social, and environmental), which is needed to understand the potential implications of digitalization for food security, ecological sustainability, equity, and governance in food systems; all of these aspects will facilitate the transition to the DA approach.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su17136053/s1, Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist.

Author Contributions: Conceptualization, W.V.-A., D.L. and L.B.; methodology, W.V.-A., M.M. and L.V.; investigation, W.V.-A.; resources, W.V.-A., L.B., F.R. and C.C.; writing—original draft preparation, W.V.-A., L.B., F.R., D.L. and C.C.; writing—review and editing, W.V.-A., L.B., F.R., D.L., M.M., L.V. and C.C.; supervision, L.B. and F.R.; project administration, W.V.-A., L.B. and F.R.; funding acquisition, L.B. and F.R. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the European Union's Horizon 2020 MSCA-RISE Programme under grant agreement 872384.

Institutional Review Board Statement: Not applicable. This study is a literature review and does not involve new data collection from humans or animals.

Informed Consent Statement: Not applicable. This study is a literature review and does not involve direct participation of human subjects or analysis of identifiable personal data.

Data Availability Statement: This is a review article, and all data discussed are derived from previously published studies cited in the reference list. No new datasets were generated or analyzed during this research.

Acknowledgments: The authors thank the National Institute of Agricultural Research and the Centre for Agroecology, Water and Resilience of Coventry University for supporting this research. This project received funding from the European Union's Horizon 2020 MSCA-RISE 2019 Programme under grant agreement 872384. Thanks to the reviewers for their valuable suggestions for improving this review.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

ACAO	Cuban Association of Organic Agriculture
ACTAF	Cuban Association of Agricultural and Forestry Technicians
AKIS	Agricultural Knowledge and Innovation Systems
ALL-Organic Project	Agroecological Living Labs Organic Project
CAE	Agroecological Collective of Ecuador
CAP	Common Agricultural Policy
CAWR	Center for Agroecology, Water, and Resilience
CAPEX	Capital Expenditure
CIRAD	Center for International Cooperation in Agricultural Research for Development
CLADES	Latin American Consortium on Agroecology and Development
CONECT-e	Sharing Traditional Ecological Knowledge
DA	Digital agroecology
DT	Digital technology
E-commerce	Electronic commerce

EU	European Union
FAS	Farm Advisory Services
FaST	Farm Sustainability Tool for Nutrients
GIS	Global information system
GPS	Global positioning system
GranSol	Solidarity Farms
H2020	Horizon 2020
INIAP	National Institute of Agricultural Research
IKIAM	Amazon Regional University
IoT	Internet of Things
IPA-LAC	Participatory Research in Agroecology in Latin America and the Caribbean
LabCampesino	Peasant Laboratory
LOASFAS	Organic Law on Agrobiodiversity, Seeds, and Agroecological Promotion
LORSA	Food Sovereignty System Comprehensive Law
MAELA	Agroecological Movement of Latin America and the Caribbean
MINAGRI	Ministry of Agrarian Development and Irrigation
MSCA-RISE	Marie Skłodowska Curie Actions - Research and Innovation Staff Exchange
NGO	Nonprofit organization
PACAT	Union of Agroecological Producers and Associative Commercialization of Tungurahua
PRISMA-ScR	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
PROBIO	Ecuadorian Corporation of Biological Producers
SANE	Sustainable Agriculture Networking and Extension
SOCLA	Latin American Scientific Society of Agroecology
UNDERTREES	Understanding Ecosystem Services of Agroforestry Systems
UK	United Kingdom

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